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WORK EXPERIENCE

Machine Learning & Fullstack Engineer (Intern), KLASS Engineering and Solutions Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art vision and audio AI models (NSG-VD, D3, ReStraV, Speech-Forensics) to evaluate real-time transparency and explainability in threat detection against proprietary pipelines.
- Integrated advanced optical flow (RAFT), segmentation (SAM 2), and point-tracking algorithms (CoTracker) to analyze physical motion derivatives in video sequences for mission-critical data.
- Engineered a comprehensive data generation pipeline linking video extension models (Wan2.2), voice cloning (Chatterbox), and lip-synchronization (LatentSync) to curate a balanced 1,200-video dataset for benchmarking.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, and deployed fullstack architectures (React, Zustand, FastAPI, WebSockets) for multi-model AI endpoints.
- Designed scalable dataset curation workflows with custom validation splits and lightweight sanity-check protocols before full-scale benchmarking.

Student Research Engineer, Singapore Institute of Technology (SIT) Aug 2024 – May 2025

- Designed and implemented an experimental temperature chamber to simulate human body heat signatures and extreme physical environments for environmental robotics testing.
- Engineered a closed-loop, thermistor-based control system to regulate precise environmental variables, ensuring high-fidelity data collection.
- Established repeatable testing and data collection protocols for faculty-led embodied AI and robotics hardware projects.

Volunteer / Full Time Research Assistant, Singapore University of Technology & Design (SUTD) - ROAR Lab Aug 2022 – May 2024

- Developed perception software and processed 3D LiDAR data for autonomous mobile robots, designing architectures integrating Embedded C++, ROS, and sensor fusion.
- Built a specialized predictive safety model to detect a stair-climbing robot's fall risk, deploying Long Short-Term Memory networks (LSTMs) to improve physical safety metrics by 50%.
- Led the design and development of an intelligent lighting control system for an autonomous vehicle, integrating it seamlessly with the main navigation system via RTOS and APIs.

Part Time/Full Time Technology Instructor, Nullspace Group Nov 2016 – Present

- Instructed students aged 5-17 in STEM-focused courses covering computational thinking, programming (C++, Python), robotics, and microcontrollers.
- Achieved MOE (Ministry of Education) Instructor Certification at age 18 (Valid Oct 2018 – Oct 2026).
- Selected to conduct the Computer Enrichment Program (CEP) at GEP schools in Singapore, completing over 1000+ teaching hours.

Robotics Engineer Intern, SOLUSTAR May 2022 - Aug 2022

- Designed and debugged PCB boards and circuitry for a disinfectant robot.
- Performed maintenance and developed custom software (Marlin) on 3D printers, improving reliability by 50% and reducing failure rates.
- Troubleshot Linux software issues for the VAL2 robot.

Robotics Instructor & Curriculum Developer, VM Education/Playfacto Jul 2020 - Dec 2020

- Pioneered and designed the new coding curriculum for the MODI (Luxrobo) platform in Singapore, publishing 5 projects of varying difficulties.
- Facilitated the training of future MODI instructors, providing solutions to potential difficulties.

PROJECTS

1. Capstone (MET4004) — Quiet Quadcopter for Indoor Flight — Project Lead (MHA)

- Led a [team size]-person team through literature review, design, and bench and flight testing, managing schedule risk and coordinating with [MHA / sponsor] and academic supervisors, [result: e.g. milestones met on

schedule].

- Designed a two-stage test campaign (a six-state factorial bench test isolating each noise-reduction mechanism, plus a flight validation stage) meeting FYP requirements while generating data for a conference abstract targeted for early 2027.
- Led a PRISMA-ScR scoping review of 33 sources on drone noise reduction to guide design decisions.
- Identified that underestimated duct and battery mass cost roughly 3 dB at hover, redirecting design effort toward mass reduction.

2. Autonomous UAV Navigation in GPS-Denied Environments

- Formulated engineering architectures for indoor UAV navigation utilizing Visual-Inertial Odometry (VIO), spatial AI depth cameras (Luxonis OAK-D Pro), and 1D Time-of-Flight sensors mapped across Raspberry Pi and Pixhawk/ArduPilot controllers.

3. Aeroacoustic Stealth for Unmanned Systems

- Investigated noise reduction strategies for UAVs—combining toroidal propellers, biomimetic owl-wing serrated blade designs, low-KV motors, and Field Oriented Control ESCs—to suppress acoustic signatures.

4. Autonomous Trolley Collection Robot

- Developed an autonomous robot for collecting stray trolleys using an STM32F7 microcontroller with RTOS, implementing tasks for obstacle detection, joystick control, and a custom serial communication protocol.

5. Custom 3D Printer Development

- Built a custom 3D printer, incorporating off-the-shelf modules and developing custom firmware in C++ (Marlin), increasing reliability and accuracy by 50%.

6. AI-Generated Video & Audio Forensics Benchmarking Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art vision and audio AI models (NSG-VD, D3, ReStraV, Speech-Forensics) to evaluate real-time transparency and explainability in threat detection against proprietary pipelines.
- Integrated advanced optical flow (RAFT), segmentation (SAM 2), and point-tracking algorithms (CoTracker) to analyze physical motion derivatives in video sequences for mission-critical data.

7. Synthetic Multimodal Deepfake Dataset Pipeline Sep 2025 – Aug 2026

- Engineered a comprehensive data generation pipeline linking video extension models (Wan2.2), voice cloning (Chatterbox), and lip-synchronization (LatentSync) to curate a balanced 1,200-video dataset for benchmarking.

8. Multi-Model AI Endpoint Platform Sep 2025 – Aug 2026

- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, and deployed fullstack architectures (React, Zustand, FastAPI, WebSockets) for multi-model AI endpoints.

9. Stair-Climbing Robot Fall-Risk Prediction (LSTM) Aug 2022 – May 2024

- Built a specialized predictive safety model to detect a stair-climbing robot's fall risk, deploying Long Short-Term Memory networks (LSTMs) to improve physical safety metrics by 50%.

10. Autonomous Vehicle Intelligent Lighting System Aug 2022 – May 2024

- Led the design and development of an intelligent lighting control system for an autonomous vehicle, integrating it seamlessly with the main navigation system via RTOS and APIs.

11. Experimental Thermal Chamber for Embodied AI Testing Aug 2024 – May 2025

- Designed and implemented an experimental temperature chamber to simulate human body heat signatures and extreme physical environments for environmental robotics testing.
- Engineered a closed-loop, thermistor-based control system to regulate precise environmental variables, ensuring high-fidelity data collection.

EDUCATION

DigiPen Institute of Technology & Singapore Institute of Technology Aug 2023 – May 2027

- B.Eng. Mechatronics Systems
- Relevant Coursework: Mechanics, Control Systems, Robotics, Embedded Systems, Machine Learning, Systems and Software Engineering

Ngee Ann Polytechnic Sep 2022 – Sep 2023

- Specialist Diploma in Robotics Engineering (SGInnovate PowerX)
- Relevant Coursework: Autonomous Robot Navigation, Machine Learning/Deep Learning for Robotics, Computer Vision

Singapore Polytechnic Apr 2017 – Sep 2020

- Diploma in Mechatronics and Robotics
- Internship grade of B+; coursework in Microcontroller Applications, PLCs, and Robotics Integration

SKILLS / COMPETENCIES

AI, Vision & Audio Analytics

1. Deep Learning
2. Computer Vision
3. Deepfake Detection
4. Generative AI
5. OpenCV
6. Optical Flow (RAFT)
7. Segmentation (SAM 2)
8. Point-Tracking (CoTracker)

Machine Learning Frameworks

1. PyTorch
2. TensorFlow
3. Keras
4. Hugging Face Transformers
5. CNNs
6. DNNs
7. LSTMs
8. YOLO
9. Vision Transformers
10. Reinforcement Learning
11. Model Benchmarking
12. Zero-shot Learning

Robotics & Autonomous Systems

1. Embodied AI
2. ROS
3. SLAM (3D LiDAR)
4. Visual-Inertial Odometry (VIO)
5. Sensor Fusion
6. RTOS
7. PID Control
8. UAV Aeroacoustics
9. PX4 Autopilot
10. ArduPilot
11. Pixhawk

Software & Fullstack

1. Python (NumPy, Pandas)
2. C++17
3. C#
4. Java
5. Embedded C
6. TypeScript
7. React
8. Zustand
9. FastAPI
10. OpenAPI
11. WebSockets
12. Git
13. Linux/Ubuntu

14. WSL

Infrastructure & Cloud Tech

1. SLURM
2. GPU Computing (CUDA)
3. NVIDIA HPC Clusters (H200 nodes)
4. Docker
5. Dev Containers
6. ONNX Runtime
7. uv
8. IoT Integrations
9. Agile/Scrum Methodologies

AWARDS / ACHIEVEMENTS / CERTIFICATES

Award / Certificate	Date
MOE (Ministry of Education) Instructor Certification (Achieved at age 18)	Valid Oct 2018 – Oct 2026
Google Crash Course on Python	
Google The Bits and Bytes of Computer Networking	
Coursera Machine Learning	

VOLUNTEERING & COMMUNITY SERVICE

ROSCon 2025 Singapore (Intrinsic / Open Robotics)

- Supported event operations, contributing to attendee experience, workshops, and logistics.

SG Makers Against COVID-19

- Initiated a pandemic response project to 3D-print ear savers for SingHealth and the local community, distributing over 10,000 units (Featured on Channel News Asia).